

EtherCAT Servo Joint Drive

STM32H743VIH6 · LAN9253 · iC-MU200 ×2 · RS-485 torque sensor
16 kHz FOC · DC-synchronised · Dual-channel STO

HARDWARE	Frozen — pin assignment is a fixed input to this design, not a variable
CONTROL RATES	16 kHz PWM + FOC · 4 kHz motion · 1 kHz supervisory · SYNC0-locked
MODULATION	Centre-aligned SVPWM, hardware-triggered synchronous sampling
FEEDBACK	iC-MU200 motor side, iC-MU200 load side, RS-485 joint torque sensor
FIELDBUS	LAN9253, distributed clock, PWM timer disciplined to SYNC0 by hardware capture
PROTECTION	Four tiers. No hardware overcurrent trip available — see §5
SUPERSEDES	Final Architecture (22 Aug), ADC and Deadline Addendum (25 Aug), Memory Layout (23 Aug), Pinout Review (25 Aug)
DATE	26 August 2026

The central design idea. Every per-period activity that hardware can do, hardware does. TIM1 drives a slaved sequencer that pulses both encoder latch lines, starts both encoder reads by DMA, and triggers the converters — all without the CPU. Only when every input is resident does a single interrupt fire, and that interrupt is the FOC ISR.

The consequence is that **one interrupt occurs per PWM period on the control path**, the ISR never waits for anything, every sample has a fixed and known relationship to every other, and the CPU keeps roughly two thirds of its capacity for the fieldbus and the application.

One principle is qualified. With the pinout frozen, both timer break inputs are committed to Safe Torque Off and no current signal reaches a comparator, so there is no sub-microsecond hardware overcurrent trip. §5 sets out what replaces it, how much slower it is, and what that costs. This is a declared limitation with a derating consequence, not a solved problem.

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1 Architectural Principles

Seven rules govern every decision below. Where two conflict, the lower-numbered wins.

#	PRINCIPLE	CONSEQUENCE
P1	Fail safe as fast as the silicon allows	Protection is tiered by response time, with the fastest tier as close to hardware as the pinout permits. Software is never the only tier. \$5 states exactly how fast that is on this board.
P2	Hardware triggers, software computes	The CPU never decides <i>when</i> to sample. Timers trigger acquisition; the CPU is invoked only once the data exists. This removes jitter and removes waiting.
P3	One time reference	PWM, both encoders, all converters, protection sampling and the fieldbus derive from or are locked to TIM1. Every sample has a known, fixed relationship to every other.
P4	Single writer per field	Each shared datum has exactly one writing context. Coherent multi-word sets cross domains by double buffer with a single-word swap, never field by field.
P5	Bounded work in bounded contexts	No blocking, no polling loops, no dynamic allocation above the main loop. Every ISR has a computable worst case.
P6	Verifiable in stages	The system builds and runs deliberately incomplete, in a defined order, each stage with a pass criterion. This is the debugging strategy, not a tutorial.
P7	Degradations are declared, not absorbed	Where the hardware prevents the intended mechanism, the architecture names the shortfall, quantifies it, and records it as an accepted risk — rather than quietly substituting something weaker and calling it equivalent.

Table 1 — Architectural principles.

2 Layer Model

FIGURE 1 — SOFTWARE LAYERS

L4	Application & Fieldbus CiA 402 drive state machine, EtherCAT slave stack, object dictionary, mailbox and CoE, parameter service, diagnostics, firmware update, scope readout. Main loop and SYNCO context.
L3	Motion Control — 4 kHz, pended Position loop on the load encoder, velocity loop on the motor encoder, torque loop on the RS-485 sensor, setpoint interpolation, feedforward, trajectory limiting, brake sequencing.
L2	Torque Control — 16 kHz, one ISR, resident in instruction TCM Offset removal, two-current Clarke, Park, current PI, inverse Park, space-vector modulation, dead-time compensation, angle advance, tier-2 checks, scope logging.
L1	Hardware Abstraction The only layer that knows pin numbers, converter channels, timer instances, scaling constants, watchdog thresholds, encoder frame formats and the fieldbus register map. Exposes a narrow verb set upward.
L0	Silicon Seven timers, three converters with analog watchdogs, three SPI, two UART, two DMA controllers, NVIC, MPU, Cortex-M7 with FPU. No trigonometric accelerator on this part — trigonometry is a table in data TCM.

The rule that gives this structure its value: L2 and L3 contain no register access and no pin numbers. Everything device-specific lives in L1. Enforce it with a build check that fails if peripheral symbols appear outside the abstraction directory.

The corollary is a genuinely useful test. **L2 and L3 should compile and run on a host computer with L1 stubbed out.** If they do, the boundary is real and you have acquired the ability to regression-test control logic against a simulated plant without hardware. If they do not, the compiler errors are an exhaustive list of your layer violations.

3 Resource Map

The hardware is fixed, so this table is the contract between the architecture and the board. Every later section refers back to it.

3.1 Timers

TIMER	ROLE	BEHAVIOUR
TIM1	Carrier and master	Centre-aligned, 16 kHz, preloaded compare and reload registers. Three complementary phase outputs with dead time. Both break inputs carry STO, latching, with no automatic output re-enable. Its trigger output slaves TIM2, TIM3 and TIM4; a second trigger output fires the converters. It raises no interrupt at all.
TIM2	Protection sampling	Slaved to TIM1, restarts every period. Triggers the converters' regular groups twenty times per PWM period so the analog watchdogs see the phase currents at 320 kHz.
TIM3	Fieldbus phase detector	Slaved to TIM1. Input capture on SYNC0. The captured count is the phase error, latched in hardware at the signal edge with no interrupt latency in it.
TIM4	Acquisition sequencer	Slaved to TIM1, restarts every period. Two channels pulse the encoder latch lines; two more generate DMA requests that start the encoder reads. Four compare values place the whole acquisition in time.
TIM5	Free-running clock	32-bit, never written by software. Serves three purposes: period-delta detection, sample timestamps, and the age tag on encoder buffers.
TIM6	Deadline monitor	Reloaded by the FOC ISR on every entry. If the ISR stops, this expires instead. Its handler sits at the highest priority.
TIM15	Brake	PWM output: full duty to pull in, reduced duty to hold. Substantially less heat than a DC hold.

Table 2 — Timer allocation. One timer remains spare.

3.2 Converters

CONVERTER	INJECTED GROUP — CONTROL	REGULAR GROUP — PROTECTION	NOTE
ADC1	Phase B, then bus voltage	Phase B, watchdog armed	Master of the dual pair. Its end-of-sequence flag invokes the FOC ISR.
ADC2	Phase C, then a repeat	Phase C, watchdog armed	Slave. Phase C is sampled at exactly the same instant as Phase B.
ADC3	Phase A	Phase A, watchdog armed	Independent converter, cannot be lockstepped with the pair.

THE SPLIT THAT MAKES THIS WORK

The injected group serves control. The regular group serves protection. The injected group fires once per PWM period at a precisely placed instant and produces the currents the FOC ISR consumes. The regular group runs continuously under analog-watchdog supervision and produces nothing the CPU ever reads — its only output is an interrupt when a threshold is crossed. Two jobs, two mechanisms, and neither interferes with the other.

3.3 Serial

INTERFACE	CONNECTED TO	ROLE
SPI — motor	iC-MU200, motor side	Commutation angle and motor velocity. Read every PWM period, DMA into a double buffer.
SPI — load	iC-MU200, load side	True joint position. Read every period alongside the motor encoder, on independent hardware.
SPI — fieldbus	LAN9253	Cyclic process data and mailbox traffic, under a single-owner state machine.
UART, RS-485	Joint torque sensor	Actual output torque. Third feedback path, motion layer only.
UART	Console	Development only. Off every deadline path.

4 The Acquisition Chain — How One PWM Period Works

This is the core mechanism of the design and the part most worth understanding precisely. The sequence below happens every $62.5\ \mu\text{s}$, and the CPU participates only in the last third of it.

4.1 Why there is no timer interrupt

A common structure is: the timer interrupt fires, the ISR starts the converters, the ISR then waits for or is re-entered on conversion complete, and only then computes. That design puts conversion latency inside the control path and needs two interrupts to do one job.

Instead, **TIM1 raises no interrupt**. It drives the converters through an internal trigger, in hardware. The converters convert. When the sequence finishes, *the converter* raises the interrupt, and that handler is the FOC ISR. The consequence is structural: **the ISR cannot begin before its input data exists, because the data existing is what caused it to begin**. There is no code path where the handler runs and the data is absent, and therefore no readiness check to write.

4.2 The carrier and the trigger points

FIGURE 2 — ONE CENTRE-ALIGNED CARRIER PERIOD

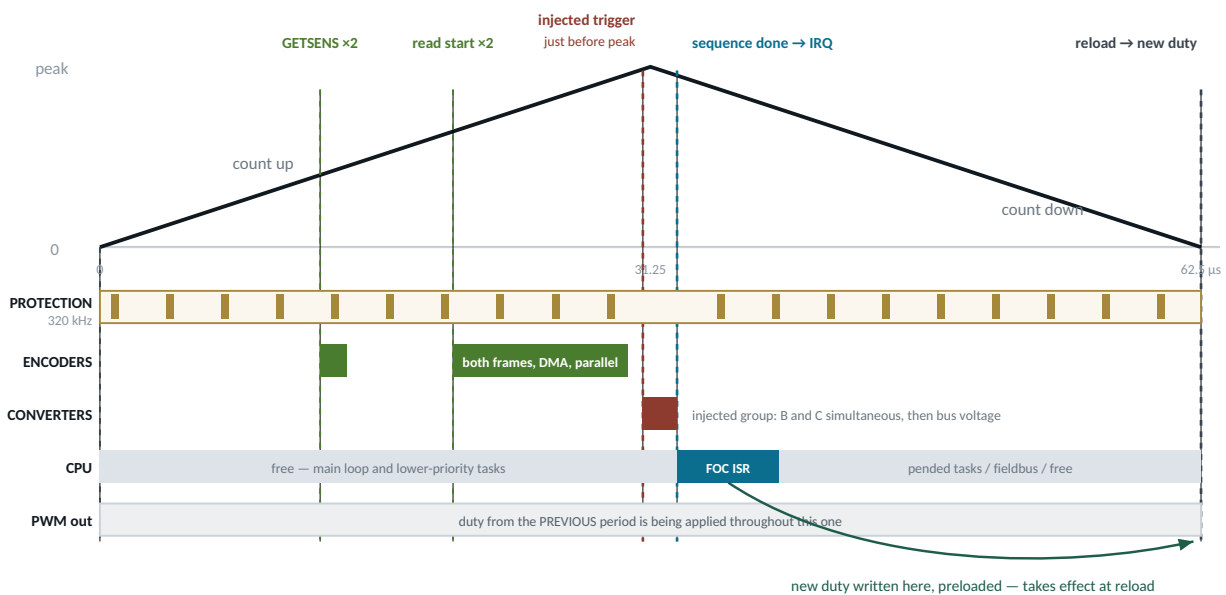


Figure 2 — Trigger instants are staggered so that the slower activity starts earlier and everything finishes together. The protection lane shows the phase currents being converted continuously at 320 kHz under analog-watchdog supervision, phased so it never collides with the injected trigger. Times are illustrative.

4.3 Step by step

#	TRIGGER	ACTOR	WHAT HAPPENS
1	TIM4, first compare	Timer → pin	A latch pulse to the motor encoder. The sensor captures its position at that instant. Because a timer generates the pulse, the latch instant is defined by silicon and has no software jitter .
2	TIM4, second compare	Timer → pin	The same for the load encoder.
3	TIM4, third compare	DMA	Starts the motor encoder read. Placed after the sensor's acquisition time so the data is ready to clock out.
4	TIM4, fourth compare	DMA	Starts the load encoder read, on independent hardware, in parallel with the first.
5	TIM1, second trigger output	Converters	The injected groups convert. Phases B and C are sampled simultaneously by the dual pair; the bus voltage follows. Results land in the data registers with no DMA and no CPU.
6	Sequence complete	CPU	The FOC ISR . Every input is already resident. It confirms the encoder transfers completed and their status lines are healthy, takes one copy of the buffer pointer, and runs the control chain. It writes the new duty cycles and finishes.
7	TIM1 reload	Timer	The preloaded compare values take effect. The new duty applies for the following period.

Table 3 — Steps 1 to 5 involve no CPU whatsoever.

WHY THE STAGGER MATTERS

An encoder frame takes several microseconds; a converter sequence takes one or two. Starting them together would mean the ISR fires on conversion completion while the encoders are still clocking. By starting the encoders earlier — placing their compare values from the measured frame duration plus margin — everything finishes before the ISR needs it, and **the encoders need no interrupt of their own**. Their completion flags and status lines are simply read as levels inside the ISR.

This is what reduces the fast path to a single interrupt per period. It also means the status lines are used as *checks* rather than as *triggers* — using them as triggers would put software latency back into an acquisition that has just been made deterministic.

4.4 Why the control path uses only two currents

Phase A sits on the independent converter and cannot be locked in step with B and C. Rather than carry that as a dependency, **the control path is built on the guaranteed-simultaneous pair alone**. For a three-wire motor these two carry all the information there is: the third current is determined by the other two.

Phase A therefore serves a single purpose in the control path — a plausibility check that the three currents sum to zero, which detects a failed sensor, a saturated amplifier, or a ground fault. Its timing relationship to the other two is irrelevant for that. The result is **zero skew where it matters and full sensor redundancy where it is cheap**, which is a better outcome than a compromise spread across three channels.

5 Protection Architecture

5.1 What the hardware permits, stated plainly

Both timer break inputs carry Safe Torque Off. No comparator-capable input receives a current signal. There is no gate-driver fault line in the pin assignment. **A phase short or shoot-through therefore has no path to the gates that does not involve the CPU.** The sub-microsecond tier the architecture originally specified does not exist on this board.

The response is not to pretend otherwise. It is to build the fastest software-mediated path the silicon allows, quantify it, and derate accordingly.

5.2 The tiers

T0	Gate driver internal — under a microsecond — now load-bearing Desaturation or drain-source monitoring, shoot-through interlock, undervoltage lockout, driver thermal shutdown. Autonomous within the driver. With the comparator tier absent, this is the only sub-microsecond protection in the system. Its presence and thresholds must be confirmed from the driver datasheet and recorded.
T1'	Oversampled analog watchdogs — about three microseconds All three phase currents converted at 320 kHz by the regular groups, each converter's analog watchdog armed with high and low limits. A crossing raises an interrupt whose handler immediately generates a software break and asserts both STO channels. The replacement for the missing tier, roughly three times slower.
T2	FOC ISR, every period — 62.5 microseconds Current sum check, filtered overcurrent, bus over- and undervoltage, over-speed, encoder status and staleness, accumulated thermal loading, missed-period detection.
T3	Supervisory, 1 kHz — milliseconds to seconds Temperatures, fieldbus watchdog, transmission cross-check, torque cross-check, parameter plausibility, fault statistics, watchdog refresh.

5.3 How the replacement tier works

A dedicated timer, slaved to the carrier, triggers a single-channel conversion on each converter twenty times per PWM period. Nothing reads the results. Each converter's analog watchdog compares the value against programmed limits and raises an interrupt only on a crossing. Worst-case detection is one sample interval plus a conversion plus interrupt entry — about three and a half microseconds.

Two properties of this arrangement are worth stating explicitly. First, **all three phases are watched**: a shoot-through circulates within a single leg, so monitoring one phase would leave the other two blind. Second, the thresholds are software-programmable, so they can be tightened during commissioning without hardware change.

PHASE THE PROTECTION SAMPLING SO IT NEVER COLLIDES WITH THE INJECTED TRIGGER

If a protection conversion is in flight when the injected trigger arrives, the injected start is delayed by up to one conversion — putting jitter back into the sampling instant that §4 exists to make deterministic.

Because both timers are slaved to the same carrier, the two are phase-locked and **the collision is avoidable by construction**. Place the injected trigger slightly before the carrier peak so that it falls cleanly in the gap between two protection samples — still well inside the zero vector, and now guaranteed never to contend. **Verify it on a scope at bring-up stage 1.**

5.4 What this costs

- **The power stage must be derated.** A trip at three and a half microseconds instead of under one means more energy passes during a fault. Compute the let-through energy for a hard short at the measured trip latency and confirm the devices survive it. This is a power-electronics calculation and it belongs to that discipline, not to firmware.
- **The gate driver's own protection must be confirmed.** If it has no desaturation or overcurrent detection, the system's fastest trip is the software-mediated one with no backup beneath it. That should be escalated rather than absorbed.
- **The watchdog interrupt shares a vector with the control ISR** for two of the three phases. Its flags are therefore checked first, before anything else in that handler. In the worst case a crossing on those phases is serviced when the control ISR finishes — still an order of magnitude better than one PWM period.

5.5 The safe state

Every fault path from every tier converges on the same terminal state:

1. Break latched by a software break event, so all outputs enter the same state a hardware break would produce and recovery requires the same explicit action.
2. Both STO channels asserted — the independent path, not dependent on the timer.
3. **All controller integrators reset.** Omit this and re-enable trips instantly, because the current loop restarts with a wound-up integrator

demanding full voltage into a stationary rotor. It presents as a hardware fault and is not one.

4. Holding brake engaged, sequenced so torque is removed only after the brake holds.
5. Fault cause latched to a status word, exposed as process data, and appended to the battery-backed fault history.
6. Scope buffer frozen, preserving the half second before the event.
7. Re-enable only by explicit command through the drive state machine. Never self-clearing.

6 Interrupt Architecture and Schedulability

6.1 How many interrupts actually fire

The concern that many interrupts will crowd each other out is worth answering with arithmetic rather than intuition. Counting per 62.5 μs period:

SOURCE	PER PERIOD	DURATION	NOTES
Sequence complete → FOC ISR	1.0	3 – 5 μs	The only guaranteed one.
Motion task (pending)	0.25	5 – 10 μs	One period in four. Now includes the torque loop.
Supervisory task (pending)	0.06	5 – 15 μs	One period in sixteen.
Fieldbus sync capture	0.06	2 – 3 μs	Phase update, starts the process-data cycle.
Fieldbus DMA complete	~0.2	< 1 μs	State machine advance and buffer swap only.
Encoder transfers	0	—	Eliminated by the stagger — flags are polled.
Encoder status lines	0	—	Configured as plain inputs. Four interrupt sources removed for identical diagnostic coverage.
Carrier timer	0	—	Not used. It drives the converters in hardware.
Fault paths	0	< 2 μs	Watchdogs, break, deadline monitor. Never in normal operation.

Table 4 — Average of roughly 1.5 interrupts per period.

Worst case is the one-in-sixteen coincidence period, when the control ISR, the motion task, the supervisory task and the fieldbus sync all land together: roughly 33 μs of 62.5 μs , about **53% utilisation**, with everything else at zero. Typical periods are nearer 10%. This is not a crowded system.

6.2 Why low priority does not mean starved

Priority answers one question only: **when two interrupts want the CPU at the same instant, which one runs first?** It says nothing about how much CPU a task receives. A pending interrupt stays pending — the controller latches the bit and services it as soon as the CPU drops below its priority level. Nothing is lost.

So the supervisory task at the bottom of the table is not starved. It is *deferred* — by at most the execution time of everything above it, which is around 22 μs in the worst case, against a 1 ms period. It has more than forty times the margin it needs.

PRIO	SOURCE	DEADLINE	WORST-CASE INTERFERENCE FROM ABOVE
0	Break — STO trip	immediate	None. Hardware already acted; this only records.
0	Phase A watchdog	immediate	Its own vector, so the fastest available path.
0	Deadline monitor	immediate	Must be able to preempt a control ISR stuck in a loop. This is why it is at the top.
1	FOC ISR (+ phase B/C watchdogs)	62.5 μs	Only the priority-0 fault paths, which are ~never. Effectively uninterruptible.
2	reserved	—	Held free for a future latency-critical need.
3	Fieldbus sync capture	~100 μs	$\leq 5 \mu\text{s}$
4	Motion task	250 μs	$\leq 13 \mu\text{s}$
5	Fieldbus DMA complete	~100 μs	$\leq 21 \mu\text{s}$
6	Supervisory task	1 ms	$\leq 22 \mu\text{s}$ against 1000 μs available
7+	Fieldbus events, STO monitors, torque sensor	none	Best effort
—	Main loop	none	Whatever remains, which is most of it

Table 5 — Priority follows rate: the faster deadline gets the higher priority. Every task's interference is far below its deadline.

THE TWO THINGS THAT WILL ACTUALLY HURT YOU — NEITHER IS INTERRUPT COUNT

Critical sections. Any interrupt disable or raised priority mask delays the control ISR by its full duration regardless of priority — **and delays the fast protection path with it.** Vendor libraries and fieldbus stacks contain long ones. Audit every one on the fieldbus path and bound your own to a handful of instructions. This, not interrupt count, is what breaks hard deadlines.

Vendor interrupt dispatch. Routing the control ISR through a generated framework handler means it walks every flag and evaluates callbacks before your code runs — hundreds of cycles of *variable* overhead on your hardest deadline. Write the handler directly and let the framework own initialisation only.

7 Deadline Failure Handling

An unfinished conversion is not observable inside the control ISR, because completion is what invokes it. The real failures are an ISR that runs **late**, one that **misses** a period entirely, and one that **stops**. Each is caught from outside, by something whose liveness does not depend on what it is watching.

LAYER	MECHANISM	CATCHES	RESPONSE
L1	Carrier position read at ISR entry	Late entry	The data is valid, only the timing is not — complete the period normally. Count the event; fault only on repetition. The deviation doubles as a free jitter measurement.
L2	Elapsed time between entries, from the free-running clock	Missed period	Safe state immediately. The angle is stale by a full period and the previous duty was reapplied blind. For a robot joint there is no benefit to limping.
L3	Deadline monitor timer, reloaded every ISR	ISR stopped	Safe state from its own highest-priority handler.
L4	Independent watchdog, refreshed only by the supervisory task	Whole chain dead	Processor reset. The refresh proves the entire chain end to end, because the supervisory task only runs if the control ISR pended it.
L5	Gate driver protection	The consequences	Autonomous. The only layer that survives total software death.

Table 6 — Late is tolerated for one period; missed is not.

A BOARD-LEVEL DEPENDENCY THAT L4 RESTS ON

A watchdog reset puts the PWM pins into a high-impedance state. **The gate driver inputs must be held safe by board-level pull-downs** for that to be a safe outcome. This is a hardware property that firmware cannot guarantee, and the whole layer is worthless without it. Confirm it on the schematic.

8 Feedback Architecture

SOURCE	MEASURES	RATE	ROLE
Motor encoder	Rotor angle, motor speed	16 kHz	Commutation and the velocity loop. Latency-critical, hardware-latched.
Load encoder	True joint angle	4 kHz	The position loop. Includes gear error, backlash and compliance — which is precisely the point.
Torque sensor	Actual output torque	1 – 4 kHz	Torque and impedance control, collision detection. Motion layer only, never the current loop. Must tolerate a missing frame without disturbing the drive.

Table 7 — Three independent measurements of the same mechanical system.

Velocity on the motor, position on the load. The inner loop sees a rigid plant and can be made fast; the outer loop sees what the application actually cares about. Closing position directly on the load introduces the two-inertia resonance between motor and load through the transmission — address that only if measurement shows it limits you, not pre-emptively.

8.1 Angle handling

Carry the electrical angle as a fixed-point phase accumulator rather than a floating-point value. Wraparound then becomes natural integer overflow — free, with no comparison, no branch, and no rounding drift accumulating over hours of running. Angle advance becomes an integer addition, and the index into the trigonometric table is a shift.

Sample the angle **once per period**, apply the advance term **once**, and use the resulting sine and cosine pair for both the forward and the reverse transform. This makes coherence between the two structural rather than something to remember. The advance term compensates the delay between sampling the currents and the new duty taking effect; **tune its coefficient empirically** by sweeping it at constant speed and load and taking the value that minimises current.

8.2 Buffer contract

- DMA writes the inactive buffer; the completion flag is **polled** by the ISR, not interrupt-driven.
- The buffer swap is a single aligned word write, which is atomic against interrupts.
- The consumer takes **one pointer copy at ISR entry** and uses that copy throughout. Re-reading the shared pointer mid-ISR reintroduces the tear it was avoiding.
- Each buffer carries a timestamp from the free-running clock. Older than one period is a tier-2 fault.
- Encoder status lines are read as levels, as validity checks — never used as triggers.

8.3 Two independent mechanical cross-checks

Having three feedback sources gives two plausibility checks, both essentially free, both running in the supervisory task:

- **Transmission integrity.** The motor angle divided by the gear ratio must agree with the load angle, within backlash plus compliance plus margin. Detects a slipping coupling, a stripped gear or failed flexspline, either encoder drifting or losing its magnet, and swapped encoder wiring.
- **Torque path.** Torque predicted from measured current, motor constant, gear ratio and efficiency must agree with the measured joint torque. Detects a drifting current sensor, an incorrect motor constant, and gearbox degradation.

Characterise both thresholds during commissioning rather than guessing them, and **publish both running discrepancies as process data**. Their slow drift over months is a genuine predictive-maintenance signal for gear wear and magnet ageing.

9 Fieldbus Architecture and Clock Discipline

9.1 The carrier disciplined to the distributed clock

The carrier must not free-run against the network's distributed clock. There is no crystal accuracy that avoids the problem; a small frequency difference still walks a full PWM period every few seconds. The consequences are setpoint jitter, a low-frequency beat on torque, and — in a multi-joint machine — loss of the cross-axis coordination that interpolated motion depends on.

The sync signal arrives on a timer **input capture**, so the counter value is latched in hardware at the signal edge. **The captured value is the phase error directly**, free of interrupt latency — measurably better than reading a counter inside a handler, which measures phase plus however long the handler took to start.

A slow loop on that phase error trims the carrier's reload value. Three properties matter: **low authority** (the trim is a fraction of a percent — you are correcting crystal drift, not slewing the drive), **low bandwidth** (a time constant of seconds, because you are tracking temperature-driven drift, and a fast loop would inject network jitter straight into your PWM period), and **defined behaviour on signal loss** (freeze at the last trim value rather than integrating away, and trigger a controlled quick stop).

Once locked, derive the motion and supervisory rates by **counting PWM periods** rather than from separate timers. One time base, no

internal drift between your own tasks.

9.2 Buffering and peripheral ownership

With the carrier locked, producer and consumer are synchronous, so **double buffering is sufficient and preferred** — lower latency and fully deterministic. Triple buffering is the correct answer to an asynchronous boundary, and the clock discipline above removes the asynchrony rather than absorbing it.

The fieldbus interface must not be arbitrated by a busy flag: testing and setting a flag from two contexts races, and the failure appears as random dropouts under load. **Give the peripheral exactly one owner** — a small state machine advanced only from the transfer-completion interrupt, servicing a request queue. The sync event starts each cycle; the main loop only enqueues. Cyclic process data always wins over mailbox traffic, and an overrun counter tells you when the cycle time is too short for the work.

9.3 Setpoint interpolation

Setpoints arrive at the network cycle rate while the position loop runs faster. Without interpolation the loop sees a staircase, which becomes a velocity step and then a torque step at the network rate — audible, and mechanically excitatory in a geared joint. Linear interpolation between consecutive setpoints is the minimum; use commanded velocity and acceleration feedforward as well if the master provides them.

10 Memory Architecture

The regions differ in wait states, in which DMA controller can reach them, in whether they are cached, and in whether they survive a power cycle. Treating those differences as a design resource is what makes the control path deterministic.

REGION	SIZE	REACHABLE BY	CONTENTS
Instruction TCM	64 K	Core only	Control ISR, transforms, controllers, modulator, motion loop, fault handlers, fast abstraction paths, vector table. Zero wait state, never cached, deterministic rather than merely fast .
Data TCM	128 K	Core only — no DMA	Main stack , trigonometric table, control state, integrators, motion state, decoded encoder values, parameters.
D2 SRAM	288 K	Core + DMA	Every DMA buffer — both encoders, fieldbus, torque sensor. Marked non-cacheable. Same domain as the controllers, so no bridge delay.
AXI SRAM	512 K	Core + DMA	Scope buffer, firmware staging, object dictionary, general allocation.
Backup SRAM	4 K	Core	Multiturn counters, fault history, crash dump, calibration validity. Never zeroed at startup .

Table 8 — Placement policy. Anything not named here defaults to AXI SRAM.

TWO SILENT HAZARDS, SOLVED BY POLICY RATHER THAN BY VIGILANCE

The general-purpose DMA controllers cannot reach tightly-coupled memory. A buffer placed there fails quietly — no error, no transfer. Cacheable memory shared with DMA gives stale reads intermittently and under load, which is among the hardest classes of bug to diagnose.

Both are solved by a single policy: a non-cacheable region for every DMA buffer, and tightly-coupled memory for everything the core owns alone. That is set up once. The alternative — cache maintenance at every transfer site — must be got right by every future call site forever.

Verify placement in the linker map with a build check. Section attributes are silently ignored more often than you would expect.

10.1 Three things this memory buys

- **A trigonometric table replacing the accelerator this part lacks.** Around ten kilobytes in zero-wait-state data memory, accurate far beyond encoder noise, and *faster* than the hardware accelerator would have been. What looked like a limitation costs nothing.
- **An oscilloscope in RAM.** A quarter of a megabyte holds eight channels at the full control rate for half a second, written circularly at a cost of well under one percent of the period budget. The decisive property is **pre-trigger capture**: any fault freezes the buffer, so you get the half second *leading up to* every trip automatically, including the ones nobody watched. Build it early — it repays itself during current-loop tuning alone.
- **State that survives power loss.** The encoders are absolute within one revolution only, so multiturn position must be held somewhere that survives a power cycle, and flash wear rules flash out. The same region holds fault history — which survives the power cycle a field failure usually causes — and crash dumps. **If its integrity check fails, that is a fault requiring re-referencing**, not a reason to assume zero: you would otherwise report a confidently wrong joint position while both encoders read perfectly.

11 State Machines

MACHINE	CONTEXT	RESPONSIBILITY
Fieldbus state machine	Main loop	The network's own initialisation and operational states, driven by the master. Leaving the operational state must command a controlled stop , not a coast.
Drive state machine	Supervisory, 1 kHz	Not ready through to operation enabled, with quick stop and fault branches. Following the standard profile gives master and tooling compatibility for free — worth taking over a bespoke design.
Fault	All tiers	Latch the cause, enter the safe state, hold until an explicit reset. Never self-clearing.

Coupling rules. Leaving the operational state triggers a quick stop. Fieldbus watchdog expiry triggers a quick stop. Any fault from any tier enters the safe state immediately, regardless of network state. The brake engages before torque is removed, and torque is applied before the brake releases — the reverse order drops the load.

12 Data Contracts

DATUM	WRITER	READERS	MECHANISM
Encoder samples	DMA	Torque, Motion	Double buffer, single-word pointer swap, timestamped
Current reference	Motion	Torque	Single aligned word write — atomic against interrupts
Position command	Fieldbus sync	Motion	Double buffer plus interpolator state
Torque reading	Torque sensor DMA	Motion, Supervisory	Double buffer plus validity flag and age stamp
Feedback and status	Torque	Motion, Supervisory, Fieldbus	Double buffer, published at ISR exit
Gains and parameters	Supervisory	Torque, Motion	Double buffer plus swap flag — never field by field
Fault word	All tiers	Supervisory, Fieldbus	Atomic bit-set only; cleared solely by an explicit reset command

Table 9 — Cross-context data, all obeying single-writer (P4).

13 Source Tree

```
src/
├── app/           # L4 – drive profile, fieldbus stack, object dictionary,
                  # mailbox, firmware update, scope readout
├── motion/        # L3 – position, velocity, torque, interpolation,
                  # feedforward, brake sequencing
├── torque/        # L2 – control ISR, transforms, controllers, modulator,
                  # angle handling, tier-2 checks
                  # NO register access. MUST compile for a host.
├── config/        # parameters only, no logic
├──   ├── motor_params # per commissioned unit – tuned values written back here
├──   ├── control_params # gains, limits, rates, angle advance, trip thresholds
├──   └── build_config  # feature selection with guards on invalid combinations
├── hal/           # L1 – ALL device and board specifics live here
├──   ├── pwm         # carrier, trigger placement, dead time, break, safe state
├──   ├── sequencer   # acquisition timing, protection sampling, clocks, monitor
├──   ├── adc         # injected and regular groups, watchdog limits, scaling
├──   ├── encoder     # frame format, integrity, DMA, double buffer, status lines
├──   ├── fieldbus    # slave controller map, single-owner state machine, capture
├──   ├── torque_sensor # protocol, bus turnaround, validity
├──   ├── safety      # STO assert and monitor, brake, emergency kill
├──   └── linker_script # memory regions and section placement
├── test/host/     # stub the abstraction layer, run L2 and L3 on a PC
                  # against a plant model
├── tools/
├──   ├── layer check  # fail the build if peripheral symbols escape hal/
├──   └── section check # fail the build if memory placement is wrong
```

The two build checks are not optional decoration. Layer discipline that exists only in a document decays within months; the check that fails the build is what actually holds the boundary. The same applies to memory placement, where a silently ignored section attribute produces a bug that only appears under load.

14 Bring-Up Plan

STAGE	SCOPE	PASS CRITERION
0	Protection. Driver protection confirmed, fast tier armed. No motor.	Gate driver protection verified from the datasheet and, where possible, by injection. Watchdog limits set below normal current; an injected fault confirmed to reach the safe state within the measured latency, on each phase in turn . STO verified from both channels and from software assert. Nothing else begins until this passes.
1	Clocks, carrier, every trigger, converters. Motor disconnected, fixed duty.	Clock tree at target frequency. Every trigger instant scope-verified against the carrier and confirmed never to collide with protection sampling. Offsets converge. Bus voltage reads true. Dead time correct. Sample time computed from source impedance, not left at a default.
2	Encoders, synchronous acquisition, no motion.	Both angles correct and repeatable when turned by hand. Latch instants confirmed synchronous on a scope. Frame integrity monitored. Gear ratio between the two verified. Both frames complete before the converters finish , with margin.
3	Open loop, forced angle. Motor connected.	Smooth rotation in the commanded direction. Currents sinusoidal and balanced. Sign of every sensed current confirmed . Sum check quiet.
4	Current loop closed on the forced angle.	Commanded current tracked with no oscillation, well-damped step response, no windup on saturation.
5	Commutation offset calibration, then closed-loop field-oriented control.	Torque per amp at expectation. Angle advance tuned by minimising current at constant load. No degradation as speed rises — the test that proves the sampling is coherent.
6	Velocity, then position on the load encoder. Torque sensor integrated.	Disturbance rejection acceptable. No resonance excitation at the joint. Both cross-check thresholds characterised by measurement, not guessed.
7	Fieldbus: state machine, process data, sync capture, clock discipline, interpolation.	Phase loop locks and holds; error bounded across a thermal cycle. Watchdog and quick stop verified by pulling the cable with the joint under load .
8	Full system, production build.	Worst-case ISR duration measured with all instrumentation enabled. Overrun counter zero across an extended run. Every tier fault-injected. Scope buffer captures a real trip.

Table 10 — Protection is stage 0, not a later addition. Each stage's pass criterion must be met before the next begins.

15 Verification Checklist

✓	ITEM	HOW TO VERIFY
☐	Driver protection confirmed in writing	Datasheet: detection present, threshold, response time. Record it — the accepted risk in §16 depends on the answer.
☐	Fast trip latency measured	Inject a step above the limit; scope from the step to gate shutdown. Must sit within the assumed budget.
☐	Fast trip covers all three phases	Inject on each phase in turn. A single-phase blind spot is not acceptable.
☐	Let-through energy survivable	Compute for a hard short at the measured latency; confirm against device safe operating area.
☐	Protection sampling never collides	Toggle a pin from each trigger and scope both. Zero overlap, every period.
☐	The current pair is truly simultaneous	Apply a common fast ramp to both channels; converted values must agree.
☐	Exactly one control interrupt per period	Pin toggle on entry. One pulse per period; two means an extra source is enabled.
☐	Encoder latch synchronous	Scope the latch pulse against the carrier. Jitter must be zero, not small.
☐	Encoders finish before the converters	Scope both, with margin. The polled flag must never be false in normal operation.
☐	Worst-case ISR timing	Pin high on entry, low on exit. Capture the coincidence period, with all instrumentation enabled.
☐	Instruction memory gives determinism	Histogram ISR duration over a long run. The spread should be near zero — visible spread means something still executes from flash.
☐	No long critical sections	Search for interrupt masking, including in vendor and stack code. Bound each one found.
☐	Memory placement correct	Linker map check in the build. No DMA buffer in tightly-coupled memory — that fails silently at runtime.
☐	Deadline monitor fires	Disable the control interrupt; the monitor must trip. Then repeat with the ISR in a deliberate infinite loop — this is what proves its priority is right.
☐	Reset leaves the bridge off	Assert reset with the motor spinning. Board pull-downs must hold the gates safe.
☐	Angle advance tuned	Sweep the coefficient at constant speed and load; minimum current wins. Re-check at top speed.
☐	Clock discipline stable	Log phase error across a thermal cycle. Bounded, no drift, freezes correctly on signal loss.
☐	Quick stop under load	Pull the cable with the joint loaded. Controlled stop, brake engages, no drop.
☐	Integrators reset on fault	Trip, re-enable, confirm no immediate re-trip.
☐	Both cross-checks fire	Loosen a coupling for the transmission check; offset the motor constant for the torque check.
☐	Persistent state behaves	Power-cycle and confirm survival. Then corrupt it deliberately and confirm a fault, not a silent zero.

16 Accepted Risks and Open Dependencies

ID	ITEM	STATUS	DETAIL AND OWNER
R1	No hardware overcurrent trip	Accepted	The fastest trip is around three and a half microseconds instead of under one. Requires power-stage derating supported by a let-through energy calculation. Owner: power electronics, with project sign-off. This is why P1 is qualified.
D1	Gate driver protection	Open	With the comparator tier absent, this is the only sub-microsecond layer. Confirm from the datasheet. If absent, R1's severity rises materially and should be escalated rather than absorbed. Close this first — it determines how serious R1 actually is.
D2	Third converter's trigger source	Open	Confirm whether it can share the control trigger. The design does not depend on it — if unavailable, Phase A runs on the protection group alone and still serves both the sum check and the fast trip.
D3	Reset holds the bridge off	Open	Board-level pull-downs on the driver inputs. The watchdog layer is worthless without this. Schematic check.
D4	STO assert response time	Open	Some STO circuits filter or delay deliberately. Measure how quickly the software assert actually kills the gates; the software break is the primary path and STO the redundant one.
D5	Serial clock budget	Open	The three SPI interfaces share a clock branch. Confirm both encoder frames complete before the control trigger, and that the fieldbus transfer fits the remaining period, at dividers actually reachable.
R2	No trace output	Accepted	An encoder interface occupies the trace pins. Standard debug is retained. Mitigated by the in-RAM scope, which is more useful for this application in any case.
R3	Watchdog shares the control vector	Accepted	Crossings on two of three phases are serviced when the control ISR finishes. Mitigated by checking those flags first, and by giving the third phase its own highest-priority vector.

Table 11 — Open dependencies should be closed before stage 0 of bring-up.

READ THIS BEFORE SIGN-OFF

R1 is the only item here that cannot be resolved in firmware. Everything else in this document is a design choice with a known answer. R1 is a permanent property of the frozen hardware, and the correct response is **a derating decision made deliberately and recorded** — not a firmware measure assumed to be equivalent to the hardware trip it replaces.